How Does the Degree of Spreading Activation of Emotional Memories Influence the Frequency and Duration of Mind-Wandering Episodes, and Does This, in Turn, Affect Performance on a Concurrent Attentional Task?

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Abstract

This study investigates the dynamics of spreading activation in emotional memory and its effects on the frequency and duration of mind-wandering episodes, as well as their impact on performance in attention-demanding tasks, using the ACT-R cognitive architecture. By simulating a Sustained Attention to Response Task (SART) and varying the :mas parameter to adjust memory activation capacities (levels 1, 2, 3), we hypothesized that higher :mas levels would lead to more persistent negative rumination. Quantitative results significantly supported this hypothesis, indicating quicker onset of persistent negative thought loops at higher :mas levels. However, structural limitations within the modeling framework itself, such as the propensity to form inescapable rumination loops, suggest caution in fully confirming the hypothesis. These findings highlight both the potential and the challenges of using cognitive architectures to model complex mental processes and suggest a need for further refinement of the model to fully understand the mechanisms linking memory activation and rumination. **Keywords:** mind wandering; rumination; spreading activation; ACT-R; SART

Introduction

Mind wandering, when attention naturally drifts from the external environment to internal thoughts, is a popular topic in cognitive psychology. Researchers have explored its connection to emotions and cognitive task performance. Rumination, a form of mind wandering marked by repetitive negative self-referential thoughts, is linked to higher depression risks and impaired cognitive performance (Nolen-Hoeksema, 2000) (Watkins, 2008). Conversely, adaptive daydreaming, involving positive or neutral internal thoughts, usually improves mood and cognitive flexibility, aiding likelihood to return to task (Killingsworth & Gilbert, 2010) (Mooneyham & Schooler, 2013).

Recent studies have provided additional insights into the relationship between emotional valence and mind wandering. For instance, (Welhaf, Astacio, & Banks, 2024) found that negative task-unrelated thoughts (TUTs) show stronger associations with SART performance measures, with a linear increase over time, while positive TUTs did not show consistent impairment. This reinforces that positive mind-wandering is less likely to hinder task resumption. As stated by the researchers, "While we found general increases in each TUT valence with time on task, the most evident, and consistent, increases were specific to negative TUTs. Thus, the collective results suggest that not all forms of mind wandering behave similarly" (Welhaf, Mata, et al., 2024).

Additionally, (Welhaf, Astacio, & Banks, 2024) found that attentional control is differentially correlated with emotional valanced TUTs, with latent variable models indicating that executive attention had different correlations depending on the emotional content of mind wandering.

Research by (Spronken, Holland, Figner, & Dijksterhuis, 2016) on temporal focus demonstrated that thoughts about the distant past and future were more positive than thoughts about the near past and future, suggesting that future-oriented thoughts, such as returning to task, are typically associated with positive valence.

Other research highlights the role of associative memory structure in depressive thought and rumination. Individuals experiencing depression tend to possess stronger associative links between negative memories, creating densely interconnected networks of negative information. (Joormann & Gotlib, 2010) This study uses the Deese-Roediger-McDermott (DRM) paradigm and finds that depressed individuals show higher false recall and recognition rates for negative critical lures compared to non-depressed controls, supporting a mood-congruent memory bias.

By synthesizing this research, particularly the increased frequency of often negatively-valenced mind-wandering in Major Depressive Disorder (Welhaf, Mata, et al., 2024) (Killingsworth & Gilbert, 2010) and the role of dense negative associative networks (Joormann & Gotlib, 2010), this study aims to computationally investigate a potential underlying mechanism. We will utilize the ACT-R cognitive architecture to model how variations in the breadth of spreading activation within memory might directly impact the tendency towards rumination during mind-wandering episodes and affect subsequent cognitive task engagement.

Our primary research question is: Does having the capacity to maintain or activate a larger number of memory associations at once lead to deeper and more prolonged negative rumination during mind wandering, compared to those with a more limited capacity for activating such associations?

By manipulating the maximum spreading activation parameter (`:mas`) in ACT-R, we aim to test whether higher values lead to more frequent and prolonged rumination-like behavior, as hypothesized by negative memory bias research (Nolen-Hoeksema, 2000)(Koster, De Lissnyder, Derakshan, & De Raedt, 2011). This investigation has implications for understanding the connection between associative spreading and depression, as well as how goal-oriented thinking can potentially counteract depressive cycles.

Model

The ACT-R cognitive architecture is employed due to its suitability for modeling interactions among rumination, negative memory bias, and task performance. Key features include: (1) chunk-based memory representations with emotional valence, (2) production rules simulating the competition between external task focus ("attend") and internal mind-wandering ("wander") goals, and (3) memory activation mechanisms—particularly spreading activation, allowing computational exploration of how the breadth or capacity of memory activation priming (controlled by the `:mas` parameter) influences rumination persistence and depth.

Model Operation: SART and Mind Wandering The model performs a *sustained attention response task* (SART), responding to targets and withholding responses to non-targets when the "attend" goal is active. If activation of the "wander" goal surpasses "attend", a production rule shifts the model into mind-wandering. Retrieval of negative memory chunks during wandering enhances the likelihood of accessing other negative chunks via spreading activation, potentially forming a *ruminative loop*. Mind-wandering persists until a positive "return-to-task" chunk is retrieved, triggering reactivation of the "attend" goal and returning focus to the SART.

Positive Valence of Task Re-engagement The "return-to-task" chunk is assumed positively valenced, supported by research linking cognitive control to intrinsic rewards (Botvinick & Braver, 2015), enhanced well-being from refocusing (Kaplan, 1995), and positive emotional associations with task-oriented and future-directed thoughts (Killingsworth & Gilbert, 2010; Wolf & Demiray, 2019; Spronken et al., 2016).

Activation and Parameter Settings Initial activations favor the "attend" goal, ensuring initial task focus. SART responses are highly practiced (strong productions), while mind-wandering chunks have moderate base-level activation. Uniform association strengths ($S_{ji} = 0.3$) facilitate thematic rumination but allow eventual disengagement. The imaginal buffer stores the last retrieved chunk, providing working memory and activation source for subsequent chunk retrieval.

Spreading Activation and Research Question Spreading activation calculates activation boosts (S_i) as:

$$S_i = \sum_k \sum_j W_{kj} \times S_{ji}$$

Activation spreads from the imaginal buffer chunk, modulated by association strengths. High associations among negative chunks reflect dense associative networks characteristic of rumination (Joormann & Gotlib, 2010; Nolen-Hoeksema, 2000). The `:mas` parameter limits total activation spread, enabling direct computational testing of whether increased spreading activation capacity enhances negative priming and thus intensifies rumination cycles (Koster et al., 2011).

Methods

This modeling study is evaluated qualitatively by comparing simulated patterns to findings from research on sustained attention, mind wandering, and rumination. From tasks like the *sustained attention response task* (SART), it is well established that negative mind wandering or rumination often correlates with performance decrements (Killingsworth & Gilbert, 2010). While specific quantitative findings (e.g., negative correlations between task-unrelated thoughts and executive attention (Welhaf, Astacio, & Banks, 2024)) exist, this work primarily aims to simulate plausible cognitive mechanisms rather than replicate exact numerical results.

Model Simulation Procedure

Simulation Runs and Trials The ACT-R model was run five times per condition to allow averaging and an assessment of variability. Each run included 200 SART trials, with 180 target stimuli (Ö) (90%) and 20 non-target stimuli (Q̈) (10%), presented in random order. Each stimulus remained visible for two seconds, followed by a 0.5-second inter-stimulus interval, offering opportunities for mind wandering.

Parameter Settings Key model parameters (base-level decay, production utilities, initial goal activations, chunk base-levels, pairwise $S_{ji} = 0.3$) were held constant across simulations, except for the experimentally manipulated parameter. This approach ensures that observed differences between conditions stem primarily from the experimental manipulation.

Independent Measures The central manipulation was variation of the global spreading activation capacity controlled by ACT-R's `:mas` parameter, which scales the total activation that spreads from active memory chunks to associated chunks. Three levels were tested (`:mas` $= 1, 2, 3$), with the expectation that higher `:mas` values would amplify associative priming among negative memory chunks, prolonging ruminative sequences.

Dependent Measures The primary dependent measure was the average duration of negative mind-wandering episodes, calculated as the mean number of consecutive negative ("Sad") memory chunks retrieved in a single off-task period before the retrieval of the "return-to-task" chunk.This index directly reflects how persistent negative thought sequences become. Two **secondary measures** provide broader insights into the model's attention shifts and task performance. The frequency of mind wandering captures the total count of transitions from the on-task ("attend") state to the off-task ("wander") state, while standard SART performance measures (overall accuracy and mean reaction time to correct target detections) assess how these cognitive fluctuations affect task execution.

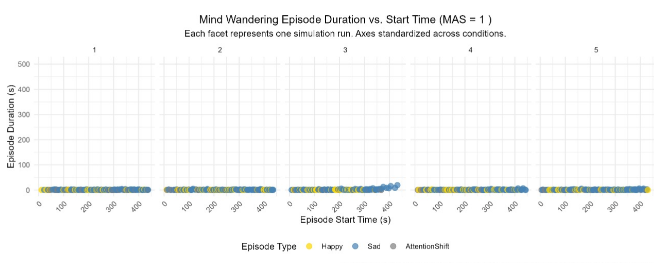
Results

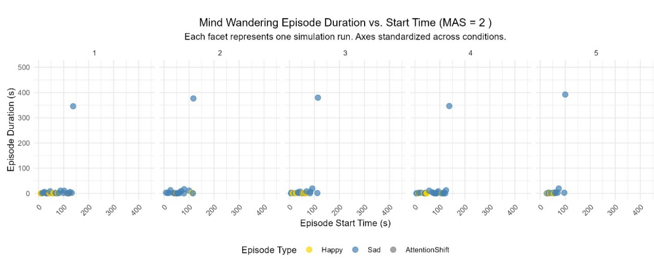
To investigate the primary research question regarding the impact of spreading activation capacity on rumination, we analyzed the output of the ACT-R model across the three experimental conditions (`:mas` = 1, 2, and 3). The analysis focused on metrics derived from the model's mind-wandering behavior during the SART simulation, primarily the average duration of negative ("Sad") mind-wandering episodes and the onset time of "infinite rumination," supplemented by visualizations of temporal dynamics.

Average Duration of Negative Mind-Wandering Episodes A one-way ANOVA was conducted to compare the average duration of "Sad" mind-wandering episodes (number of consecutive negative retrievals) across the three `:mas` conditions, as initially planned. The results indicated a statistically significant difference between the groups, F(2, 12) = 29.41, p ¡ 0.001. A Tukey HSD post-hoc analysis revealed that the average duration for the `:mas` = 3 condition was significantly longer than for both the `:mas` = 1 condition (p ¡ 0.001) and the `:mas` = 2 condition (p ¡ 0.001). The difference between `:mas` = 1 and `:mas` = 2 was not statistically significant (p = 0.893).

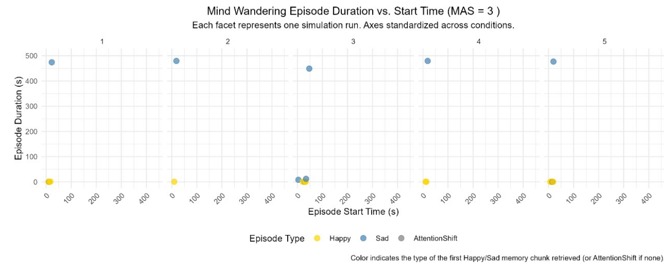
However, observation of the model's behavior revealed that this average duration metric is heavily influenced by instances where the model entered states of "infinite rumination," failing to retrieve the "return-to-task" chunk. These extremely long final episodes disproportionately inflate the calculated average, particularly in the higher `:mas` conditions where infinite rumination occurred more frequently.

Figures below illustrate the increasing prevalence of deep, unbroken sequences of negative retrievals (yellow dots) in the higher `:mas` conditions, showing how episodes leading to infinite rumination dominate the later simulation stages.





Onset Time of Infinite Rumination Given the limitations identified with the average duration metric, the onset time of infinite rumination was examined as an alternative measure reflecting the model's susceptibility to entering highly persistent ruminative states. This analysis included only the runs where infinite rumination was observed (3 out of 5 runs for

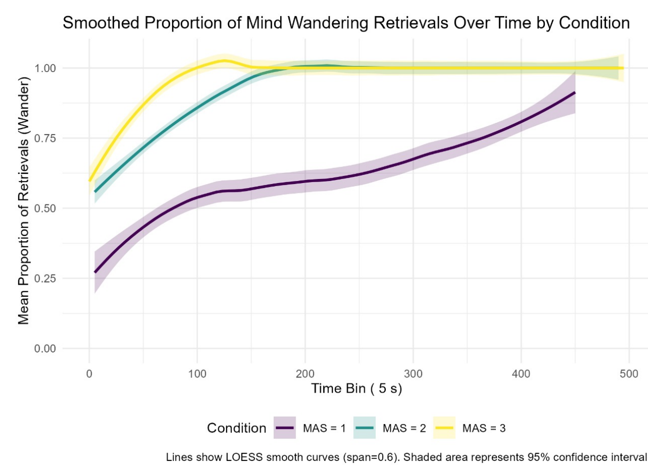


`:mas`=1; 5 out of 5 runs for both `:mas`=2 and `:mas`=3).

A one-way ANOVA performed on the onset time revealed a highly significant difference between the conditions, F(2, 10) = 904.3, p ¡ 0.001. The mean onset times indicate that infinite rumination begins significantly earlier as the `:mas` parameter increases: approximately 435.04 seconds for `:mas` = 1, 121.34 seconds for `:mas` = 2, and 25.17 seconds for `:mas` = 3. A Tukey HSD post-hoc analysis confirmed that all pairwise comparisons were statistically significant (p ¡ 0.001).

The observation that infinite rumination occurred in a higher proportion of runs at higher `:mas` levels, combined with the significantly earlier onset times, provides quantitative evidence related to the persistence of rumination influenced by spreading activation capacity.

Figure further illustrates these increase.



Note on SART Performance Data Detailed SART performance metrics (accuracy, reaction time) are not presented here. The onset of infinite rumination, particularly its early occurrence in higher `:mas` conditions, acts as a significant confound. Runs effectively terminate early with respect to task engagement when infinite rumination begins, making direct comparison of task performance over the full intended simulation duration misleading. The relationship between the earlier onset of persistent rumination and inferred task performance decrements will be elaborated upon in the Discussion section.

Discussion

This study investigated how manipulating the capacity for spreading activation (`:mas` parameter at levels 1, 2, and 3) within an ACT-R model influences simulated rumination during a Sustained Attention to Response Task (SART). The central hypothesis was that a higher capacity for spreading activation would lead to more persistent negative thought loops. Our simulation results, particularly the analysis of infinite rumination onset times, support this hypothesis, indicating that stronger associative activation makes the model more susceptible to entering and remaining in persistent, maladaptive ruminative states. While direct SART performance metrics were confounded by the early onset of infinite rumination in high `:mas` conditions, the observed increase in persistent off-task states aligns with findings where increased rumination negatively impacts task performance in humans (Nolen-Hoeksema, 2000).

Model Evaluation Evaluating the model reveals both strengths and limitations. Quantitatively, the model captures the general principle that more intense internal distraction (here, simulated rumination driven by higher `:mas`) compromises task engagement. Qualitatively, the ACT-R framework provides mechanistic transparency in how memory retrieval and goal competition lead to shifts between on-task and off-task states. The model's parameters were minimally tuned, relying on standard ACT-R principles to maintain theoretical consistency. Furthermore, the model implicitly captures mood-congruent retrieval effects through the valence-based associative structure and activation dynamics; negative thoughts preferentially prime other negative thoughts. Spreading activation was also appropriately constrained to memory chunks, preventing interference with other cognitive processes within the model architecture.

However, significant weaknesses exist. The most prominent issue is the model's tendency towards "infinite rumination loops," particularly at higher `:mas` values. While persistent rumination is characteristic of some clinical states, the model's propensity to enter inescapable negative loops likely exceeds typical human cognitive patterns. This phenomenon also rendered the initially planned primary measure, average rumination duration, problematic due to skewing by these extremely long final episodes, necessitating the shift to analyzing onset time. This suggests the parameters governing the balance between spreading activation (`:mas`) and associative strength (S_{ji}) may require refinement; lower S_{ji} values might allow for a clearer examination of `:mas` effects without immediately triggering infinite loops, potentially yielding more human-like results across a wider range of `:mas` values. Additionally, the modeling approach relies on implementing observed effects (like mood congruency) as architectural features or rules, which might limit the generalizability of conclusions drawn solely from this specific implementation. The assumption that the "return-to-task" chunk is positively valenced and readily accessible, while grounded in supporting literature, also simplifies the complex process of disengaging from rumination; the model showed that higher `:mas` primarily strengthened negative-to-negative links, making retrieval of the positive "return-to-task" chunk less frequent but not impossible, partially aligning with the difficulty of interrupting real-world rumination.

Relation to Other Research The finding that increased spreading activation capacity (`:mas`) leads to earlier onset of persistent rumination aligns with theories proposing that individuals with strong, dense associative networks among negative memories are more vulnerable to ruminative cycles that impair sustained attention (Watkins, 2008; Koster et al., 2011). Our simulation provides computational support for the crucial role of associative retrieval processes, modulated by overall activation capacity, in controlling the persistence of mind-wandering episodes, consistent with prior modeling work (Mooneyham & Schooler, 2013; van Vugt & van der Velde, 2015). Specifically, the earlier onset time observed with higher `:mas` can be interpreted as reflecting a breakdown in cognitive control, where the system becomes captured by internal associations, aligning with the idea that broader associative activation among negative concepts, potentially analogous to denser negative networks in depression (Joormann & Gotlib, 2010), facilitates persistent and difficult-to-disengage rumination. The model's behavior also resonates with findings by (Welhaf, Astacio, & Banks, 2024), which demonstrated a stronger negative impact of negative versus positive or neutral task-unrelated thoughts on SART performance, supporting the model's differentiation based on valence.

In summary, this computational study demonstrates that manipulating a single parameter (`:mas`) reflecting the capacity for associative memory priming within ACT-R significantly impacts the model's susceptibility to persistent, rumination-like thought patterns. The earlier onset of infinite rumination under higher `:mas` conditions supports the central hypothesis and suggests that enhanced associative processing among negative memories can indeed lead to more entrenched rumination. This provides a potential mechanistic link between individual differences in cognitive architecture (specifically, spreading activation capacity) and vulnerability to maladaptive cognitive states like rumination.

Future work should aim to address the model's limitations. Refining the interplay between `:mas` and S_{ji} values could mitigate the extreme infinite rumination effect and allow for investigation across a broader parameter space. Incorporating temporal aspects of memory, as research suggests temporal focus influences valence (Spronken et al., 2016), could add depth beyond simple valence tagging. Furthermore, developing a more explicit and nuanced representation of mood state, perhaps as a continuous variable influencing retrieval thresholds rather than relying solely on implicit mood congruency via activation, could enhance the model's psychological plausibility and generalization. Assigning continuous valence values to memories instead of strict positive/negative categories might also yield more realistic dynamics. By refining these aspects, future iterations of the model could provide deeper insights into the cognitive mechanisms underlying rumination and its impact on goal-directed behavior.

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